

Sowmith - GNS3

File Edit View Control Node Annotate Tools Help

Filter

ATM switch

cisco7200

Cloud

Ethernet hub

Ethernet switch

Frame Relay switch

NAT

VPCS

+ New template

case study 1:
when both pc's are in same network

Switch1

mac address table

ping 192.168.1.2

AA
PC1
VPCS
192.168.1.1/24

BB
PC2
VPCS
192.168.1.2/24

ARP table
192.168.1.2 BB

ARP table
192.168.1.1 AA

ARP request
Sender mac AA
Sender IP 192.168.1.1
Target mac FF:FF:FF
Target IP 192.168.1.2

ARP Reply
Sender mac BB
Sender IP 192.168.1.2
Target mac AA
Target IP 192.168.1.1

PC1 will generate ICMP Echo Request
SMAC AA
SIP 192.168.1.1
DMAC BB
DIP 192.168.1.2

PC2 will generate ICMP Echo Reply
SMAC BB
SIP 192.168.1.2
DMAC AA
DIP 192.168.1.1

ICMP - Internet Control Messaging Protocol
ARP - Address Request Protocol

Topology Summary

Node	Console
PC1	telnet localhost:5001
PC2	telnet localhost:5003
Switch1	none

Servers Summary

Sowmith CPU 9.6%, RAM 60.4%

Console

GNS3 management console.
Running GNS3 version 2.2.61 on Windows (64-bit) with Python 3.13.14 Qt 6.10.0 and PyQt 6.10.2.
Copyright (c) 2006-2026 GNS3 Technologies.
Use Help -> GNS3 Doctor to detect common issues.

=>

X: -313.0 Y: -223.0 Z: 2.0

```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
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Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.1/24
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.107 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=11.430 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.097 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.439 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=0.835 ms

PC1> █
```

```
Welcome to Virtual PC Simulator, version 0.6.2
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For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 192.168.1.2/24
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0

PC2> █
```

- ▶ Frame 1: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
- ▶ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
- ▼ Address Resolution Protocol (request)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: request (1)
 - Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
 - Sender IP address: 192.168.1.1
 - Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 - Target IP address: 192.168.1.2

0000	ff ff ff ff ff ff 00 50 79 66 68 00 08 06 00 01	P yfh.....
0010	08 00 06 04 00 01 00 50 79 66 68 00 c0 a8 01 01	P yfh.....
0020	ff ff ff ff ff ff c0 a8 01 02 00 00 00 00 00 00	
0030	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

No.: 1 · Time: 0.000000 · Source: Private_66:68:00 · Destination: Broadcast · Protocol: ARP · Length: 64 · Info: Who has 192.168.1.2? Tell 192.168.1.1

☒ Show packet bytes Layout: Vertical (Stacked)

Close

- ▶ Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
- ▶ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
- ▼ Address Resolution Protocol (reply)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: reply (2)
 - Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
 - Sender IP address: 192.168.1.2
 - Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
 - Target IP address: 192.168.1.1

0000	00 50 79 66 68 00 00 50 79 66 68 01 08 06 00 01	Pyfh..P yfh----
0010	08 00 06 04 00 02 00 50 79 66 68 01 c0 a8 01 02	P yfh----
0020	00 50 79 66 68 00 c0 a8 01 01 00 00 00 00 00 00	Pyfh.....
0030	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

No.: 2 · Time: 0.001510 · Source: Private_66:68:01 · Destination: Private_66:68:00 · Protocol: ARP · Length: 64 · Info: 192.168.1.2 is at 00:50:79:66:68:01

- ▶ Frame 3: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
- ▶ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:01 (00:50:79:66:68:01)
- ▼ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.2
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 84
 - Identification: 0xf2d3 (62163)
 - ▶ 000. = Flags: 0x0
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 64
 - Protocol: ICMP (1)
 - Header Checksum: 0x0482 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.168.1.1
 - Destination Address: 192.168.1.2
 - [Stream index: 0]
- ▼ Internet Control Message Protocol
 - Type: Echo (ping) request (8)
 - Code: 0
 - Checksum: 0x4c18 [correct]
 - [Checksum Status: Good]
 - Identifier (BE): 54258 (0xd3f2)
 - Identifier (LE): 62163 (0xf2d3)
 - Sequence Number (BE): 1 (0x0001)
 - Sequence Number (LE): 256 (0x0100)
 - [Response frame: 4]
 - ▶ Data (56 bytes)

0000	00 50 79 66 68 01 00 50	79 66 68 00 08 00 45 00	.Pyfh..P yfh...E.
0010	00 54 f2 d3 00 00 40 01	04 82 c0 a8 01 01 c0 a8	.T....@.....
0020	01 02 08 00 4c 18 d3 f2	00 01 08 09 0a 0b 0c 0d	L.....
0030	0e 0f 10 11 12 13 14 15	16 17 18 19 1a 1b 1c 1d	
0040	1e 1f 20 21 22 23 24 25	26 27 28 29 2a 2b 2c 2d	..!""#\$% &'()*+,-
0050	2e 2f 30 31 32 33 34 35	36 37 38 39 3a 3b 3c 3d	./012345 6789:;<=
0060	3e 3f		>?

No.: 3 · Time: 0.015823 · Source: 192.168.1.1 · Destination: 192.168.1.2 · Protocol: ICMP · Length: 98 · Info: Echo (ping) request id=0xd3f2, seq=1/256, ttl=64 (reply in 4)

☒ Show packet bytes

Layout: Vertical (Stacked) ▼

- ▶ Frame 4: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
- ▶ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
- ▼ Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.1.1
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 84
 - Identification: 0xf2d3 (62163)
 - ▶ 000. = Flags: 0x0
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 64
 - Protocol: ICMP (1)
 - Header Checksum: 0x0482 [validation disabled]
 - [Header checksum status: Unverified]
 - Source Address: 192.168.1.2
 - Destination Address: 192.168.1.1
 - [Stream index: 0]
- ▼ Internet Control Message Protocol
 - Type: Echo (ping) reply (0)
 - Code: 0
 - Checksum: 0x5418 [correct]
 - [Checksum Status: Good]
 - Identifier (BE): 54258 (0xd3f2)
 - Identifier (LE): 62163 (0xf2d3)
 - Sequence Number (BE): 1 (0x0001)
 - Sequence Number (LE): 256 (0x0100)
 - [Request frame: 3]
 - [Response time: 0.860 ms]
 - ▶ Data (56 bytes)

0000	00 50 79 66 68 00 00 50	79 66 68 01 08 00 45 00	.Pyfh..P yfh...E.
0010	00 54 f2 d3 00 00 40 01	04 82 c0 a8 01 02 c0 a8	.T....@.
0020	01 01 00 00 54 18 d3 f2	00 01 08 09 0a 0b 0c 0d	T... ..
0030	0e 0f 10 11 12 13 14 15	16 17 18 19 1a 1b 1c 1d	
0040	1e 1f 20 21 22 23 24 25	26 27 28 29 2a 2b 2c 2d	.. !"#\$\$% &'()*+,-
0050	2e 2f 30 31 32 33 34 35	36 37 38 39 3a 3b 3c 3d	./012345 6789;<=
0060	3e 3f		>?

No.: 4 · Time: 0.016683 · Source: 192.168.1.2 · Destination: 192.168.1.1 · Protocol: ICMP · Length: 98 · Info: Echo (ping) reply id=0xd3f2, seq=1/256, ttl=64 (request in 3)

☒ Show packet bytes Layout: Vertical (Stacked) ▼